

Intestinal nematode infections

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For 3rd year HO students

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Intestinal nematodes

- adult range from few cms to 30cm or more – often visible
- female and male worms-separate, with females often larger
- infection acquired either by ingestion of eggs/larvae or larval penetration of intact skin
- Life cycle-simple/direct. Some with lung migration

- some may be transmitted from person to person, some require development in **soil** prior infection

egg – L1 – L2 – L3 – L4 – Adult

Different life cycle stages can differ dramatically in their metabolism (aerobic larvae, anaerobic adults)

- most common in regions with poor fecal sanitation, particularly in resource-poor countries in the tropics and subtropics

- disease caused by **accumulation**-intensity more important, disease/morbidity- related to worm burden combined with **malnutrition**.
- In persons with marginal nutrition, intestinal helminth infections may **impair growth** and development.
- some chronic infections – chronic ill health- associated with impaired: **cognitive** development, intellectual ability, loss of memory: but reversible
- potential-stunting, impaired school performance

- **Disease mechanisms:** could be related to obstructions, competition for nutrients, inflammatory reactions , bleeding associated with iron and protein loss -----
- diagnosis: often confirmed by finding eggs /larvae in stool and other techs.
- Eosinophilia and elevated serum IgE levels are features of many helminthic infections
- Significant protective immunity to intestinal nematodes appears not to develop in humans

Intestinal nematodes infections...

- Ascariasis
- Hook worm diseases
- trichuriasis
- Enterobiasis
- strongyloidiasis

ASCARIASIS

- *A. lumbricoides* is the largest intestinal nematode parasite of humans, reaching up to 40 cm in length.
- Most infected individuals have low worm burdens and are asymptomatic.
- Clinical disease arises from larval migration in the lungs or effects of the adult worms in the intestines.

Epidemiology

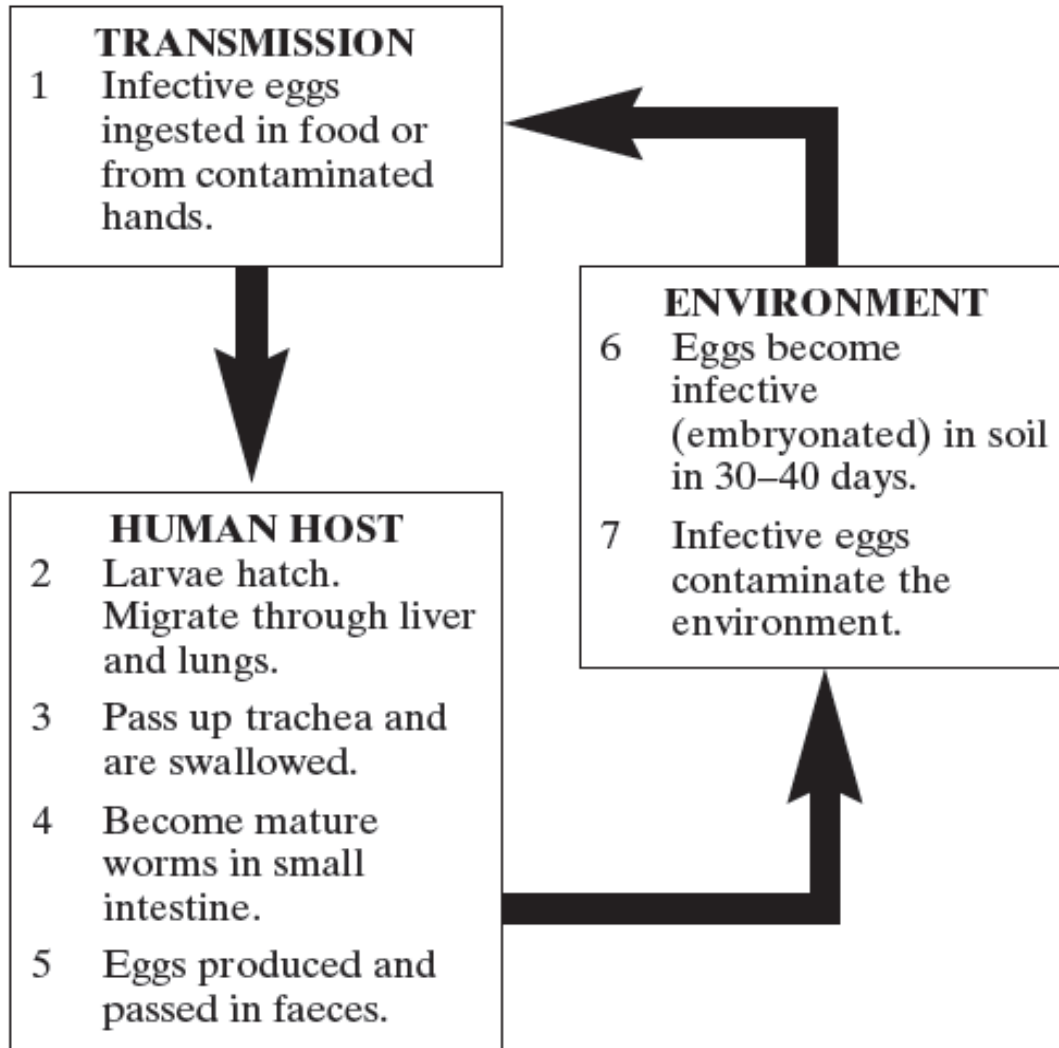
- **The most common parasite of humans where sanitation is poor.**
- School children (5-10 years of age) are most affected.
- Highly prevalent in moist tropical countries

- Transmission typically occurs through fecally contaminated soil and is due either to a lack of sanitary facilities or to the use of human feces as fertilizer.
- With their propensity for hand-to-mouth fecal carriage, **younger children** are most affected.
- Infection outside endemic areas, though uncommon, can occur when eggs on transported vegetables are ingested

Life Cycle

- **Adult worms live in the lumen of the small intestine.**
- **Mature** female *Ascaris* worms are *extraordinarily fecund*, each producing up to 240,000 eggs a day, which pass with the feces.
- Ascarid eggs, which are **remarkably** resistant to environmental stresses, become infective after **several weeks** of maturation in the soil and can remain **infective for years**.
- After infective eggs are swallowed,
- larvae hatched in the intestine invade the mucosa, migrate through the circulation to the lungs, break into the alveoli, ascend the bronchial tree, and return via swallowing to the **small intestine**, where they develop into adult worms.
- Between 2 and 3 months elapse between initial infection and egg production. Adult worms live for 1–2 years.

Life cycle



Clinical Manifestation

- Most infections go unnoticed until large worm is passed in feces and occasionally the mouth and nose.
- Migrant larvae may cause itching, wheezing and dyspnea, fever, cough bloody sputum and Eosinophilia may occur – often self-limiting
- Abdominal pain may arise from intestinal or duct (biliary, pancreatic) **obstruction**.
- Serious complications include bowel obstruction (most common among complications) due to knotted /intertwined worms
(3-5%) of cases in endemics, 17% mortality
- hepatobiliary obstructions, cholecystitis, acute cholangitis, pancreatitis and hepatic abscesses

Clinical Features

- **Lung Phase:** **5-6 days after exposure**
 - Pneumonitis (**Loeffler's pneumonia**) due to penetration of lung capillaries by juvenile parasite causing hemorrhage
- **Intestinal Phase:** **2-3 months after infection**
 - abdominal pain, diarrhea
 - malabsorption of nutrients
 - stunted growth, cognitive impairment
 - intestinal obstruction by high worm numbers
- **Wandering worms:** **cause pathogenesis:**
 - intestinal perforation: peritonitis
 - blockage of bile ducts
 - obstruction of respiratory tract
 - liver abscesses

Diagnosis

Clinical and imaging techs: ultrasonography, CT scan....often for obstructive cases

Lab

- Microscopic identification of eggs in a stool sample
- Adult worms passed from anus, mouth or nose.

Treatment

Chemotherapy

- Albendazole (400 mg once), mebendazole (500 mg once), or
- ivermectin (150–200 µg/kg once) is effective. These medications are contraindicated in pregnancy, however. Pyrantel pamoate (11 mg/kg once; maximum, 1 g) is safe in pregnancy.

Clinical – surgical intervention for – bowel obstruction

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Prevention and control

1. Treatment of cases (patients Vs mass treatment)
2. Sanitary disposal of feces
3. Prevent soil contamination in areas where children play
4. Promote good personal hygiene (hand washing

Trichuriasis

A nematode infection of the **large intestine**, usually asymptomatic in nature.

Infectious agent

Trichuris trichuria (whip worm)

Epidemiology

Occurrence- Worldwide, especially in warm moist regions.

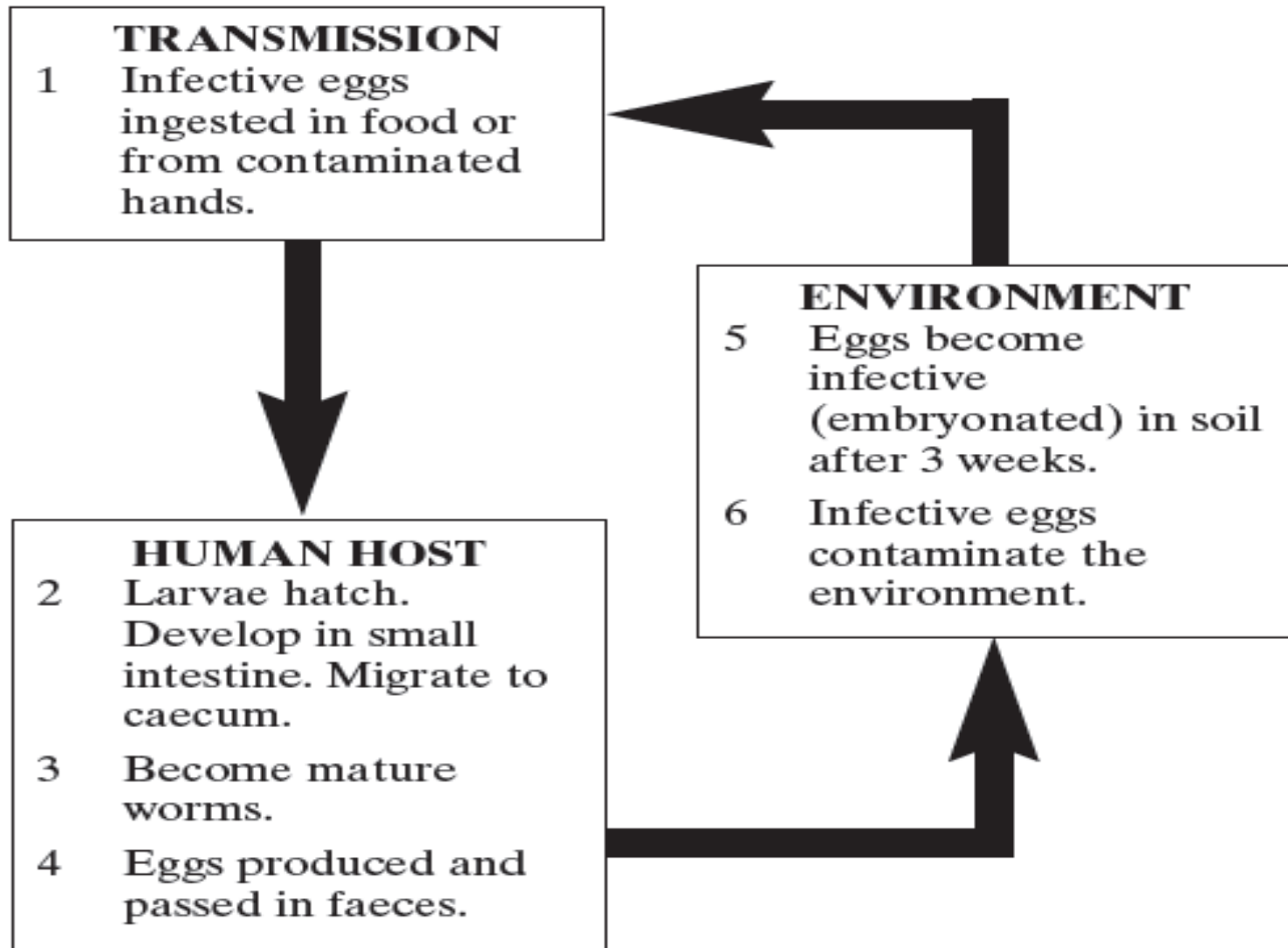
Common in children 3-11 years of age.

Reservoir- Humans

Mode of transmission- Indirect, particularly through pica or ingestion of contaminated vegetables.

Not immediately transmissible from person to person.

Life cycle



Clinical Features

- Tissue reactions to *Trichuris* are mild.
- *Most infected* individuals have no symptoms or eosinophilia.
- Heavy infections may result in abdominal pain, anorexia, and bloody or mucoid diarrhea resembling inflammatory bowel disease.
- Rectal prolapse can result from massive infections in children, who often suffer from malnourishment and other diarrheal illnesses.
- Moderately heavy *Trichuris burdens* also contribute to growth retardation.

Clinical manifestation

- Adult *Trichuris* worms reside in the colon and cecum, the anterior portions threaded into the superficial mucosa.
- Severity is directly related to the number of infecting worms.
- Most infected people are asymptomatic.
- Abdominal pain, tiredness, nausea and vomiting, diarrhea or constipation are complaints by patients.
- **Rectal prolapse** may occur in heavily infected very young children.

Prolapsed rectum with adult *Trichuris trichiura*



Diagnosis

- Demonstration of eggs in feces.

Treatment

- Mebendazole (500 mg once) or albendazole (400 mg daily for 3 doses) is safe and effective for treatment. Ivermectin (200 µg/kg daily for 3 doses) is also safe but is not quite as efficacious as the benzimidazoles.
- **Prevention and control**
 1. Sanitary disposal of feces
 2. Maintaining good personal hygiene (i.e. washing hands and vegetables and other soil contaminated foods)
 3. Cutting nails especially in children
 4. Treatment of cases.

Entrobiasis

(Oxyuriasis, pinworm infection)

Definition

- A common intestinal helminthic infection that is often asymptomatic.

Infectious agent

- *Entrobias vermicularis*

Epidemiology

- **Occurrence- Worldwide, affecting all socio-economic classes** with high rates in some areas.
- Prevalence is highest in school-aged children, followed by preschools and is lowest in adults except for mothers of infected children.
- Infection usually occurs in more than one family member.

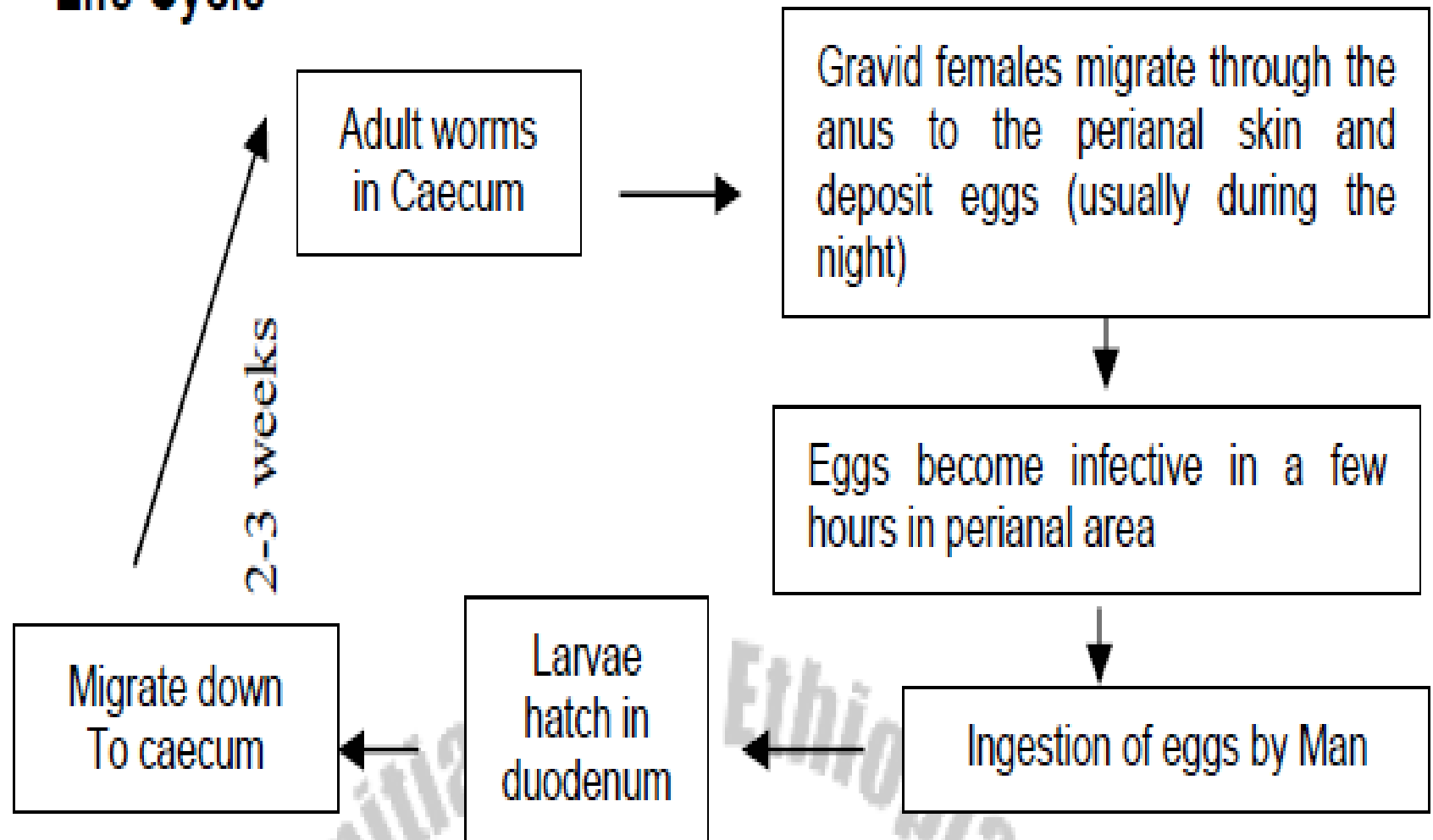
Reservoir- Human

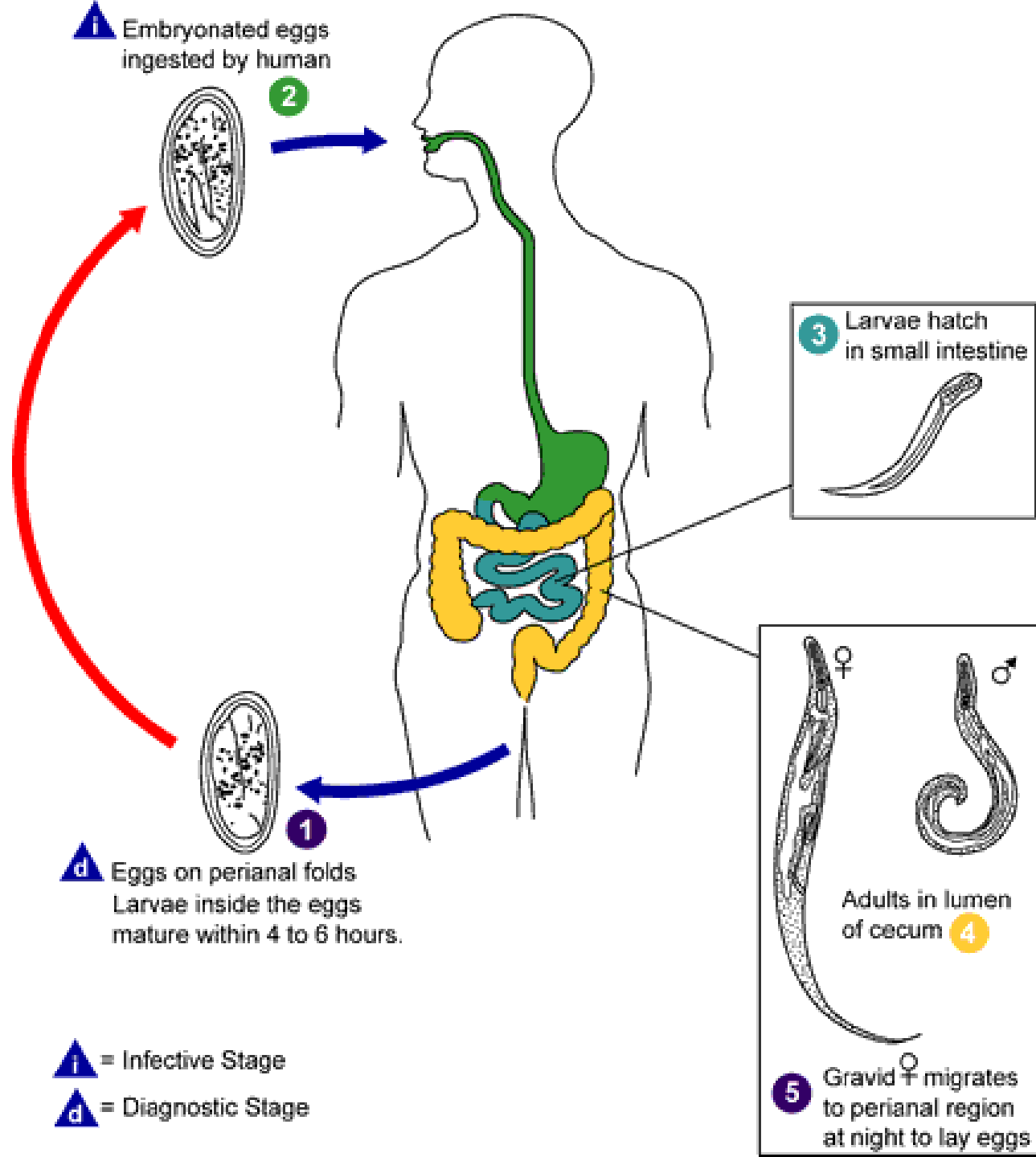
Mode of transmission- Direct transfer of infective eggs by hand from anus to mouth of the same or another person or indirectly through clothing, bedding, food or other articles contaminated with eggs of the parasite.

- Pinworms eggs can be spread throughout a house and difficult to eliminate.
- Small children are most apt to pick them up during the “teething stage.”



Life Cycle





Clinical features

- Nocturnal anal **pruritis**.
 - The cause of this is unknown, but may be related to the intensity of the infestation, and/or an allergic reaction to parasite
- Sleeplessness, because of the irritation
- Vulvovaginitis, and even urethritis may occur in girls when migrating worms lay their eggs in these sites
- Abdominal pain or appendicitis resulting from the worms are considered to be very rare

Heavy Infection of *Enterobius vermicularis*



Photo: Martin Weber, MD, Children's Hospital; Hannover Medical School; Hannover, Germany

Diagnosis

- Since pinworm eggs are not released in feces, the diagnosis cannot be made by conventional fecal ova and parasites tests.
- Instead, eggs are detected by the application of clear cellulose acetate tape to the perianal region in the morning.
- After the tape is transferred to a slide, microscopic examination will detect pinworm eggs, which are oval, measure 55 by 25 μm , and are flattened along one side.

Prevention and control

- Educate the public about hygiene (i.e. hand washing before eating or preparing food, keeping nails short and discourage nail biting).
- Treatment of cases
- Reduce overcrowding in living accommodations.
- Provide adequate toilets.
- Washing of the anal skin each morning
- Washing of clothing worn at night
- Wearing tight-fitting cotton pants to infected children

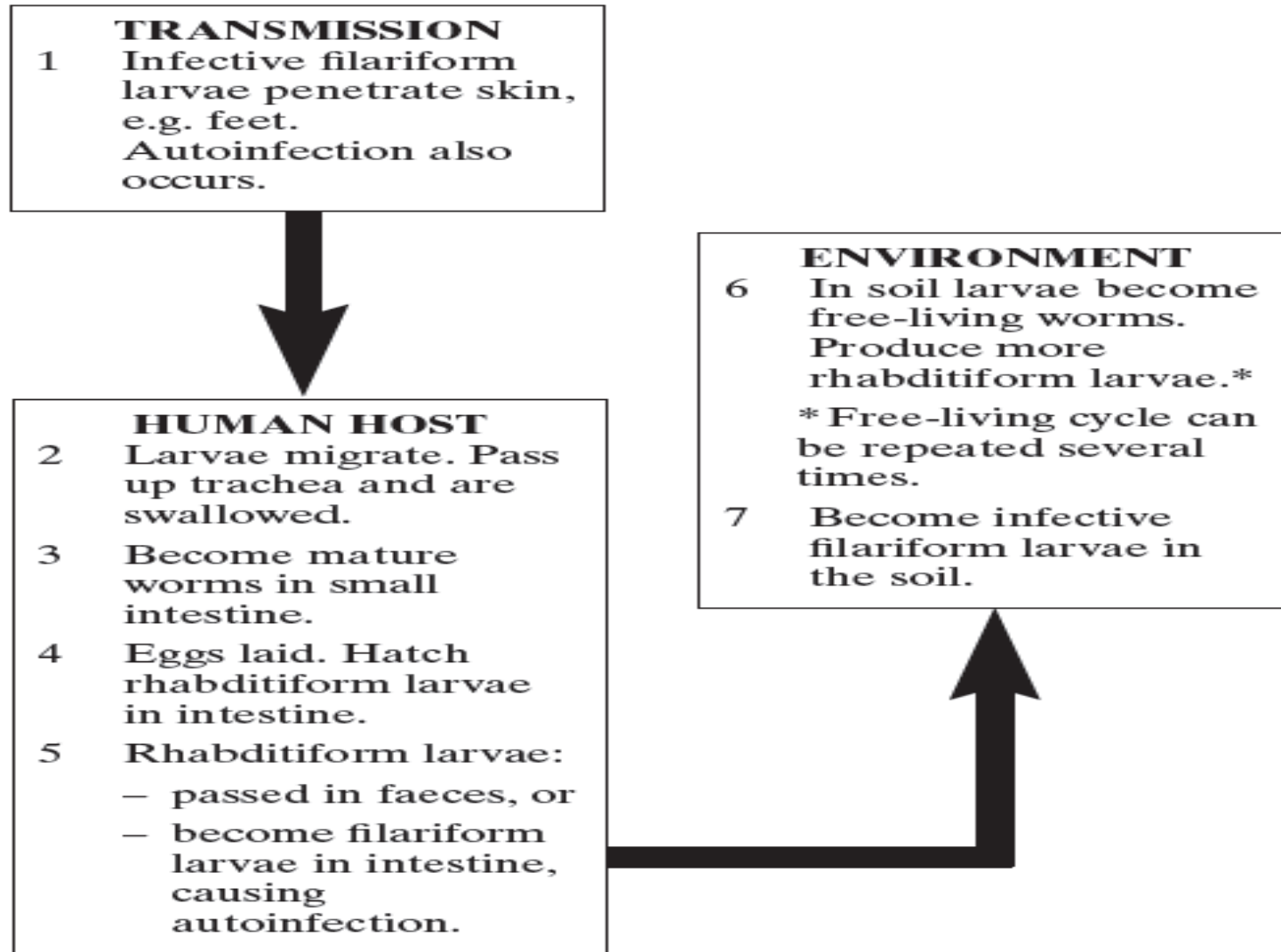
Rx

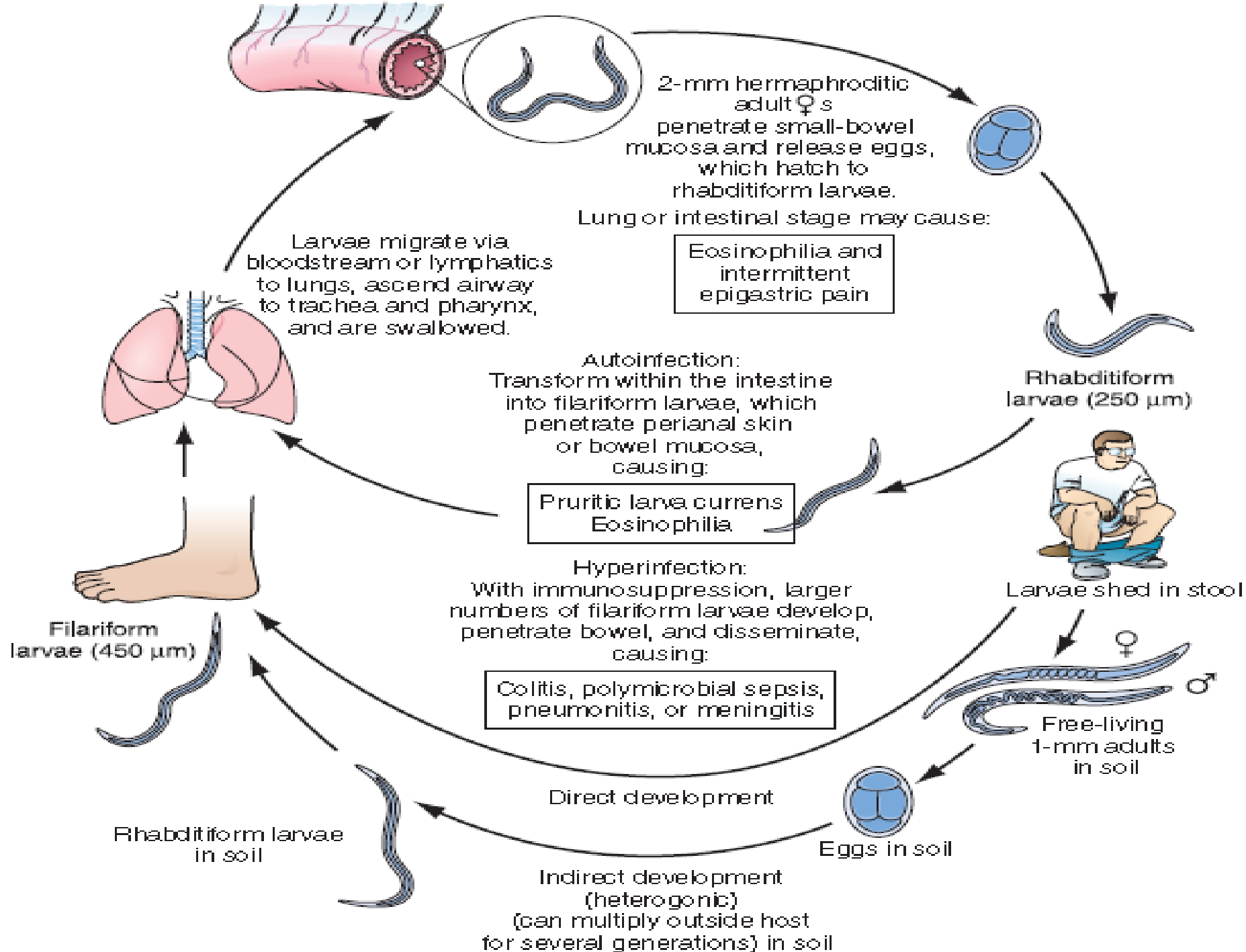
- mebendazole (100 mg once),
- albendazole (400 mg once), or
- pyrantel pamoate (11 mg/kg once; maximum, 1 g),
- Treatment of **household** members is advocated to eliminate asymptomatic reservoirs of potential reinfection.

STRONGYLOIDIASIS

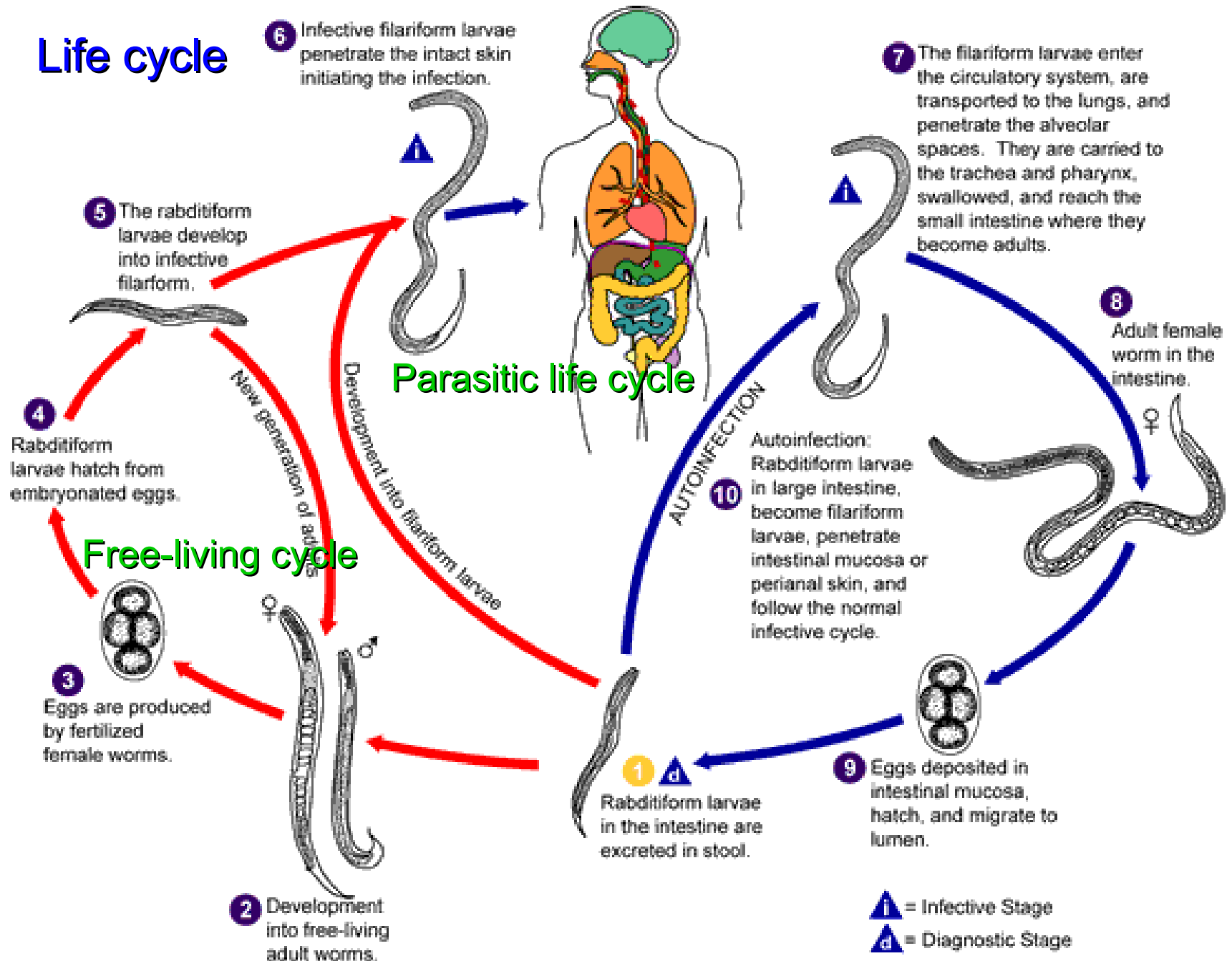
- *S. stercoralis* is distinguished by its ability—unusual among helminths— **to replicate in the human host**.
- This capacity permits ongoing cycles of autoinfection as infective larvae are internally produced.
- Strongyloidiasis can thus persist for decades without further exposure of the host to exogenous infective larvae.
- In immunocompromised hosts, large numbers of invasive *Strongyloides* larvae can disseminate widely and can be fatal.

Life cycle





Life cycle



Clinical features

- When penetrating the skin, *S. stercoralis* larvae can cause an itchy dermatitis and rash.
- During migration of the larvae, allergic and respiratory symptoms may occur.
- Most infections are without serious symptoms. Heavy infections (especially common in children) can cause dysentery, malabsorption, steatorrhoea, and dehydration with electrolyte disturbance.
- Abdominal pain is common and occasionally finger clubbing. There is usually an eosinophilia.

- ***Hyperinfection:*** Autoinfection with *S. stercoralis* can become overwhelming and sometimes fatal when the body's normal immune responses are reduced, e.g. by drugs, steroids, malnutrition, pregnancy or puerperium, or when other diseases are present which cause immunosuppression.
- In such infections, larvae can be found in most tissues and serous cavities of the body.
- Ulcerative enterocolitis can occur with dysentery or rectal bleeding. It is therefore important to check for strongyloidiasis in those at risk of developing disseminated hyperinfection. Gram-negative sepsis, pneumonia, or meningitis may complicate or dominate the clinical course.

Diagnosis

- Identification of larvae in stool specimen. Or duodenal aspirate and biopsy

Prevention and control

1. Proper disposal of human excreta (feces)
2. Personal hygiene including use of footwear.
3. Case treatment.

Rx

- Even in the asymptomatic state, strongyloidiasis must be treated because of the potential for subsequent fatal hyperinfection.
- Ivermectin (200 µg/kg daily for 2 days) is more effective than albendazole (400 mg daily for 3 days).
- For **disseminated** strongyloidiasis, treatment with ivermectin should be extended for at least 5–7 days or until the parasites are eradicated

Hookworm disease

(Ancylostomiasis, Necatoriasis)

Definition

- A common chronic parasitic infection with a variety of symptoms usually in proportion of the degree of anemia

Infectious agent

- *Ancylostoma duodenale* and
- *Necator americanus*

Epidemiology

- **Occurrence- Widely endemic in tropical and subtropical** countries where sanitary disposal of human feces is not practiced and the soil moisture and temperature conditions favor development of infective larvae.

Reservoir- Humans

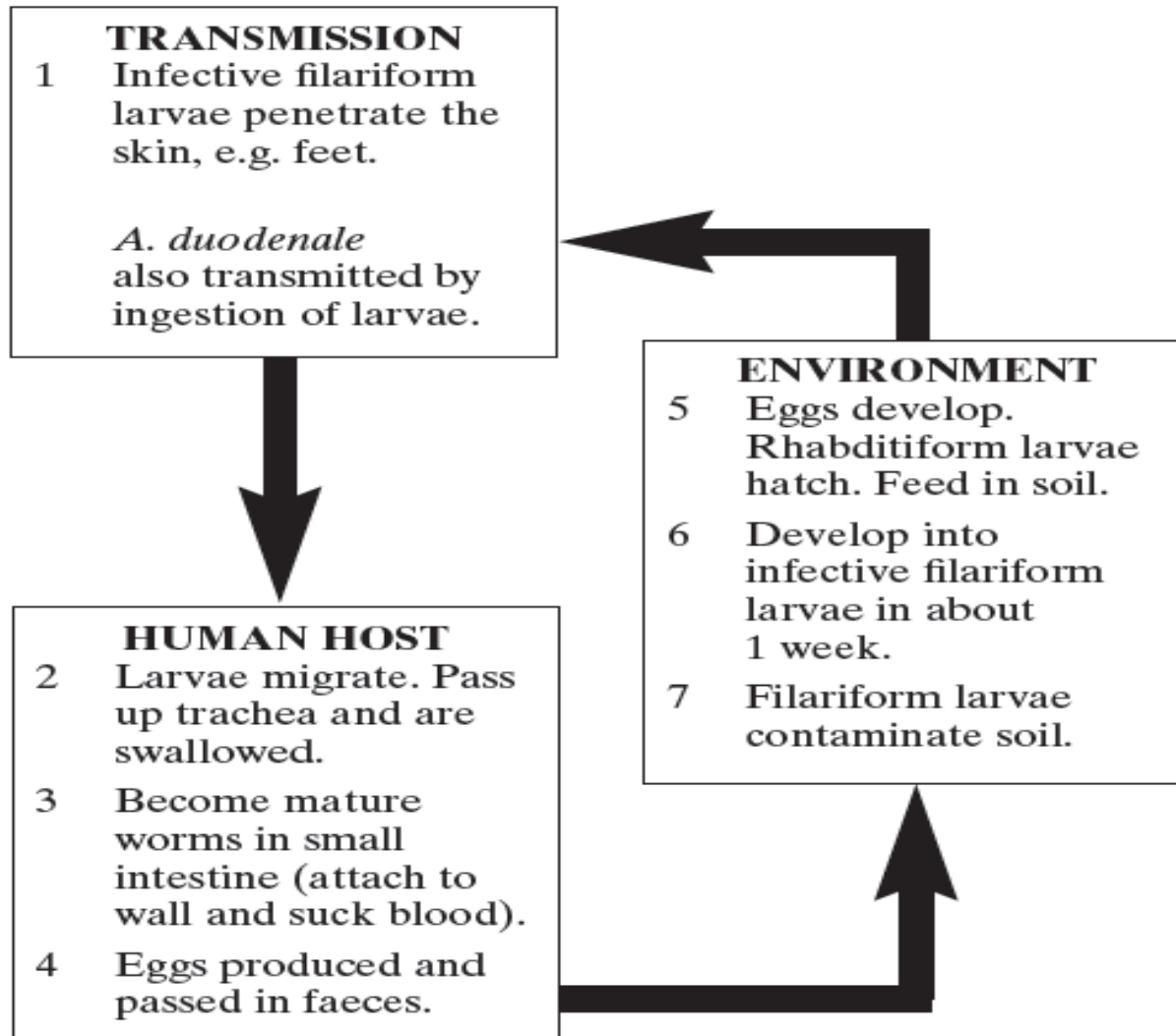
Mode of transmission- Through skin penetration by the infective larvae.

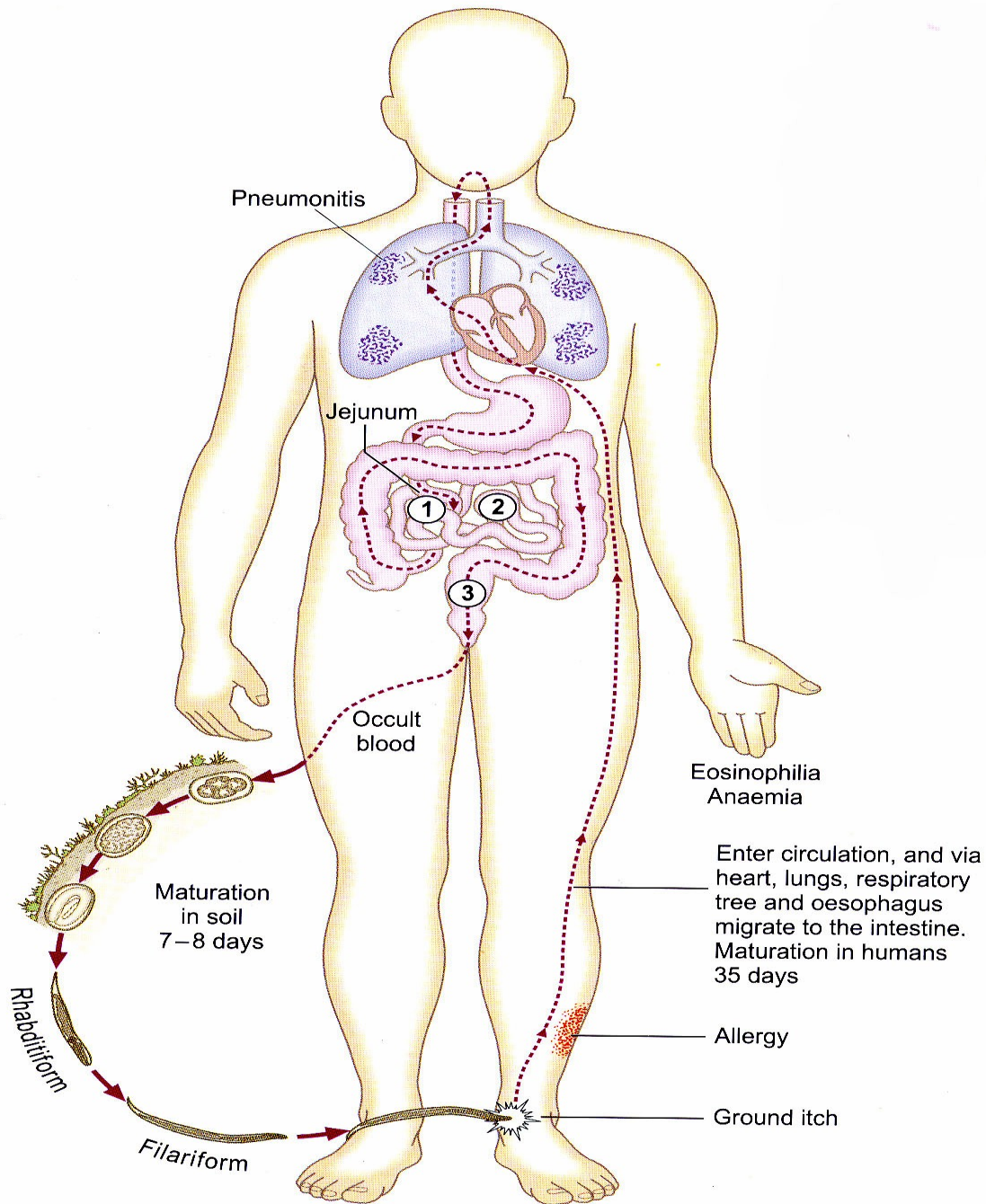
Incubation period- Symptoms may develop after a few weeks to many months depending on intensity of infection and iron intake of the host.

Period of communicability- Infected people can contaminate the soil for several years in the absence of treatment.

Susceptibility and resistance- Susceptibility is universal. No evidence that immunity develops with infection.

Life cycle





Pathogenesis

- attachment to intestine – blood loss and protein losing enteropathy
- production of coagulation inhibitory proteins

Clinical Manifestation

- The clinical manifestation is related to:
 1. Larval migration of the skin Produces transient, localized maculopapular rash associated with itching called ground itch.
Invasion of the skin is mediated by production of proteolytic enzymes
 2. Migration of larva to the lungs. Produces cough, wheezing and transient pneumonitis.
 3. Blood sucking
 - Light infection-no symptoms
 - Heavy infection-result in symptoms of peptic ulcer disease like epigastric pain and tenderness. Further loss of blood leads to anemia manifested by exertional dyspnea, weakness and light-headedness.

Hookworms

- The main concern with hook worm disease is blood loss
- 0.03 ml (N.a.) to 0.26 ml (A.d) per worm,
- up to 200 ml per day in heavy infections
- Chronic heavy infection results in anemia and iron deficiency
- Together with malnutrition infection can severely stunt growth and development in children
- Anemia leads to weakness and fatigue in adults



Clinical features of hookworm disease

Site	Symptoms	Pathogenesis
Dermal	Local erythema, macules, papules (ground itch)	Cutaneous invasion and subcutaneous migration of larva
Pulmonary	Bronchitis, pneumonitis and, sometimes, eosinophilia	Migration of larvae through lung, bronchi, and trachea
Gastro-intestinal	Anorexia, epigastric pain and gastro-intestinal hemorrhage	Attachment of adult worms and injury to upper intestinal mucosa
Hematologic	Iron deficiency, anemia, hypoproteinemia, edema, cardiac failure	Intestinal blood loss

Diagnosis

- Demonstration of eggs in stool specimen.

Treatment

1. Mebendazole or
2. Albendazole or
3. Levamisole

Prevention and control

1. Sanitary disposal of feces
2. Wearing of shoes
3. Case treatment.

- Hookworm infection can be eradicated with several safe and highly effective anthelmintic drugs, including albendazole (400 mg once), mebendazole (500 mg once), and pyrantel pamoate (11 mg/kg for 3 days).
- Mild iron-deficiency anemia can often be treated with oral iron alone.
- Severe hookworm disease with protein loss and malabsorption necessitates nutritional support and oral iron replacement along with deworming.